

1. Consider two harmonic waves (having general form: $A \cos(kx - \omega t)$) are travelling simultaneously along a wire given by

$$W_1 = 0.001 \cos(4.0x - 200t), \quad \text{and} \quad W_2 = 0.001 \cos(3.8x - 180t).$$

Here x is in meter, t is in sec and ω is in Hz. Find the phase and group velocities of the resultant wave.

2. An atom can radiate at any time after it is excited. It is found that in a typical case, the excited atom has a life time of about 10^{-8} sec. This means that during this period it emits a photon and gets de-excited. What is the minimum uncertainty in the frequency of the photon emitted? Calculate the uncertainty in the energy of the excited state of the atom, ΔE .
3. Use the uncertainty relation to estimate the ground state energy (minimum possible energy) of a one dimensional harmonic oscillator having potential $V = \frac{1}{2}m\omega^2 x^2$.
4. Consider a particle whose wave function is given by,

$$\begin{aligned} \psi(x) &= Nx e^{-\alpha x}, & x > 0, \\ &= 0 & x < 0, \end{aligned}$$

where α and N are positive real constants.

- (a) Find the normalized wave function.
- (b) For what value of x does the probability of finding the particle be maximum?
- (c) What is the probability that the particle is found between $x = 0$ and $x = 1/\alpha$?
- (d) Calculate the expectation values of x and x^2 and hence the uncertainty Δx .
- (e) Calculate the expectation values of p and p^2 and hence the uncertainty Δp .
- (f) Check HUP.

[Useful relation:

$$\int_0^\infty x^n e^{-ax} dx = \begin{cases} \frac{\Gamma(n+1)}{a^{n+1}} & \text{for } n > -1, \operatorname{Re}(a) > 0, \\ \frac{n!}{a^{n+1}} & \text{for } n = 0, 1, 2, \dots, \operatorname{Re}(a) > 0. \end{cases}$$

5. Consider $\psi(x, t)$ as a solution of the Schrodinger equation for a free particle of mass m in one dimension, and

$$\psi(x, 0) = A \exp(-x^2/a^2).$$

- (a) At time $t = 0$, find the probability amplitude in momentum space.
- (b) Find $\psi(x, t)$.

6. A particle of mass m is confined to a one-dimensional box (*i.e.* infinite potential well) of width a , *i.e.* $0 \leq x \leq a$. At $t = 0$, its normalized wave function is

$$\psi(x, t = 0) = \sqrt{8/5a}[1 + \cos(\pi x/a)] \sin(\pi x/a).$$

- (a) What is the wave function at a later time $t = t_0$?
- (b) What is the average energy of the system at $t = 0$ and at $t = t_0$?
- (c) What is the probability that the particle is found in the left half of the box (*i.e.*, in the region $0 \leq x \leq a/2$) at $t = t_0$?